

Cloud Computing

Unit – 2

❖ Cloud Computing Fundamentals:

- **Motivation for Cloud Computing:**

- 1. Cost Efficiency (Operational vs. Capital Expenditure):**

- Before cloud computing, users and enterprises had to make a substantial upfront investment in computing resources—hardware, software, networking, storage—along with the maintenance costs. These resources had to be bought, housed, maintained, and operated on-premises, leading to high capital expenditures.
- Cloud computing shifts this burden to a pay-per-use or subscription model, which reduces costs to operational expenditures. Enterprises or individuals only pay for the resources they consume, making it much more economical.

- 2. Elimination of Large-Scale Investment:**

- For organizations needing vast computing power, investing in physical infrastructure was an enormous financial strain. Cloud computing allows these organizations to scale their infrastructure up or down as needed, without the massive upfront costs of purchasing and maintaining expensive equipment.

- 3. Scalability and Flexibility:**

- Cloud computing offers the flexibility to increase or decrease computing resources based on demand. This dynamic scaling ensures that users only pay for what they need, avoiding over-investment in unnecessary infrastructure.

- 4. Improved Access to Computing Resources:**

- Cloud computing eliminates the need for users to own and manage their own computing infrastructure. Whether for individuals or businesses, cloud computing provides easy and convenient access to computing resources when needed. This can be compared to how electricity is consumed, where users only pay for the amount they use, without worrying about generating or maintaining the resource.

- 5. Security and Data Protection:**

- A key benefit of cloud computing is data security. Even if a user loses a laptop or experiences a system failure, their data remains safe and secure because it is stored remotely on the cloud provider's infrastructure. This adds an additional

layer of protection and peace of mind for users, as they do not need to worry about data loss from local device failures.

6. Increased Adoption by Small and Medium Enterprises (SMEs):

- Cloud computing has become especially popular among small and medium-sized enterprises (SMEs) that may not have the resources to invest in and manage large-scale infrastructure. Cloud services allow these businesses to access computing resources that were previously out of reach, giving them the flexibility to grow and scale without significant capital investment.

7. No Need for New Infrastructure or Personnel:

- Cloud computing allows organizations to expand their computing capabilities without the need to invest in new physical infrastructure or hire additional staff. This eliminates both capital investment and the training costs typically associated with maintaining a large IT system.

The cloud computing is motivated by the need to make computing more affordable, scalable, and accessible. It allows users to avoid high upfront costs, reduce infrastructure management burdens, enhance data security, and meet computing demands on the fly. This makes cloud computing a highly attractive and practical solution for individuals, enterprises, and SMEs alike.

• The Need of Cloud Computing:

1. Convenience and Accessibility

- In the past, to access files or documents, people had to use physical storage devices such as USB flash drives, external hard drives, or CDs. Cloud computing removes the need for these devices because files saved in the cloud can be accessed from any device, such as a computer, smartphone, or tablet, using an internet connection.
- Whether you're at home, at work, or traveling, the cloud ensures that your files are readily available, making it highly convenient for users who need to access or update their documents from different locations.

2. Enhanced Collaboration and Sharing

- Cloud computing facilitates easy file sharing and collaboration. Services like Dropbox allow users to store and share documents, images and other files with friends, colleagues or clients. Multiple people can work on the same file simultaneously, improving team productivity and collaboration.

- This ease of sharing and collaborating makes cloud computing indispensable for businesses and individuals who need to work together across different locations.

3. Data Security and Reliability

- Storing data in the cloud significantly reduces the risk of data loss. Unlike physical storage devices (e.g., USB drives, CDs) that can be lost, damaged, or stolen, cloud services store data securely in remote servers, often with advanced backup and redundancy systems in place.
- The likelihood of losing data on the cloud is much lower, and in the event of a hardware failure, cloud services are generally more reliable in recovering and protecting data.

4. Scalability and Flexibility

- Cloud services offer the flexibility to scale up or down based on the user's needs. This is particularly valuable for businesses that may need more storage or computing power during certain periods but don't want to make long-term investments in physical infrastructure.
- The cloud allows for on-demand use of computing resources, which is efficient and cost-effective.

5. Reduced Physical Storage Dependency

- Cloud computing minimizes the need for physical storage devices. Files can be stored in the cloud, reducing clutter and reliance on physical media. This can help individuals and businesses manage their data more effectively and securely.
- With cloud storage, users don't have to worry about running out of space on their local devices.

6. Access Control and Privacy Management

- While cloud services offer convenience and security, there's always a need for vigilance when it comes to privacy. Users must ensure they implement strong passwords and manage privacy settings effectively.
- Cloud providers typically offer various access controls to restrict who can view or modify files, giving users the ability to manage data security and privacy at a granular level.

7. Cost-Effectiveness

- Cloud computing can also be more cost-effective than traditional methods of data storage and management. Instead of purchasing and maintaining physical storage infrastructure, users can pay for cloud services based on their actual usage (e.g., pay-per-use model).

- For businesses, this reduces capital expenditure and helps avoid large upfront costs for hardware and storage systems.

The need for cloud computing is driven by its **convenience**, **reliability**, and ability to facilitate easy **access**, **collaboration** and **data security**. Cloud services ensure that files and documents are always accessible, regardless of location, while minimizing the risk of data loss. Additionally, they provide flexibility, scalability and control over data, making them a critical resource for both individuals and businesses. In a world where collaboration and access to data at any time is increasingly vital, cloud computing is an essential technology.

- **Definition of Cloud Computing:**

Cloud Computing means **storing** and **accessing the data** and **programs** on **remote servers** that are hosted on the internet instead of the **computer's hard drive or local server**. Cloud computing is also referred to as Internet-based computing, it is a technology where the resource is provided as a service through the Internet to the user. The data that is stored can be files, images, documents or any other storable document.

The following are some of the Operations that can be performed with Cloud Computing:

- Storage, backup and recovery of data
 - Delivery of software on demand
 - Development of new applications and services
 - Streaming videos and audio
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- ❖ **Importance of Cloud Computing:**

Cloud computing is important because it makes technology and data more accessible, efficient and flexible. It helps businesses and individuals save money, resources and time by providing easier access to data and software.

Example: Imagine a small business that doesn't have the budget to buy expensive computers and software. With cloud computing, they can rent software and storage space online, pay for only what they use and avoid large upfront costs.

- ❖ **Why Cloud Computing?**

Cloud computing helps with flexibility, scalability and cost-effectiveness. It allows users to access data and apps from anywhere at any time, as long as they have an internet connection. It also enables companies to grow without having to worry about managing physical infrastructure.

Example: If you are using an app like Netflix, you don't need to store all the movies on your device. Netflix stores the content on cloud servers and you can watch it from your phone, tablet or TV as long as you have an internet connection.

❖ **Uses of Cloud Computing:**

1. **Storage:** Cloud services allow you to store files, photos, and videos safely and access them anytime.

Example: Google Photos, where you can store all your pictures online and access them from any device.

2. **Software as a Service (SaaS):** Cloud computing allows you to use software applications over the internet without needing to install them on your computer.

Example: Google Docs, where you can create and edit documents online without needing Microsoft Word.

3. **Collaboration:** Cloud computing makes it easier for teams to work together on projects, no matter where they are.

Example: A team working on a document at the same time through Google Docs, without needing to send files back and forth.

4. **Backup:** Cloud computing helps keep data safe by regularly backing it up online.

Example: iCloud automatically backs up your iPhone's data, like contacts, photos, and apps so you don't lose it if you lose your phone.

- **Cloud Computing Is a Service:**

Cloud computing is essentially a service that allows users to store, access and manage data remotely over the internet. Unlike traditional methods where data is stored on physical devices (like computers or external hard drives), cloud computing offers a more flexible, accessible and secure way to manage information. Let's explore this through the example of **Google Drive**, which offers basic cloud computing services to its users.

- 1. Accessibility Anywhere, Anytime:**

- Google Drive provides users with the ability to store documents, photos, videos, and other files in the cloud. This means that you can easily access your files from any device—whether it's your phone, tablet, laptop or desktop—whenever you have an internet connection.
- For example, you might upload a report to Google Drive from your home computer, and later access or edit it from your laptop at work, or even on your phone during your commute. This eliminates the need to carry physical storage devices, and ensures your files are always available to you, no matter where you are.

- 2. Easy Sharing and Collaboration:**

- One of the key features of Google Drive is its ability to facilitate easy sharing and collaboration. You can upload files to the cloud and share them with others by simply sending them a link. This makes it incredibly easy to collaborate on documents, spreadsheets, and presentations without the need for emailing attachments or using USB drives.
- For instance, you can upload a presentation to Google Drive, share the link with your colleagues, and they can edit and provide feedback on the file in real-time, all without needing to be in the same location.

- 3. Data Security and Backup:**

- Cloud services like Google Drive offer a layer of data security by creating a backup of your files on remote servers. While it is always a good idea to keep local backups of your important files (such as on a USB flash drive or external hard drive), the risk of losing data on physical devices is much higher than with cloud services.
- With Google Drive, your files are securely stored and backed up, so even if your computer crashes or you lose access to your local storage devices, your files remain safe and accessible from anywhere. The chances of Google Drive losing your data are far less compared to the risks associated with physical storage devices, which can easily be damaged, lost, or stolen.

Key Features of Cloud Computing as a Service:

- **On-Demand Service:** Cloud computing provides access to resources (like storage, computing power, etc.) when needed, on a pay-per-use basis. With Google Drive, you can store a certain amount of data for free, and upgrade to larger storage plans as needed.
- **Scalability:** As your data needs grow, cloud services can easily scale to meet those demands. Google Drive allows you to increase your storage without any hassle.
- **Flexibility:** Users can access their data from any device connected to the internet, whether they are using a smartphone, laptop, or desktop.

Cloud computing is fundamentally a service that provides users with storage, data access, sharing, and security—all without the need for physical storage devices. Services like **Google Drive** illustrate how cloud computing can make data management more convenient, secure, and collaborative, showcasing how technology has transformed the way we store and interact with information.

• Cloud Computing Is a Platform:

Cloud computing is not only a service but also a **platform** for delivering a wide range of applications and resources over the internet. Through cloud computing, the **web** is transforming into a platform that supports applications and services, much like traditional operating systems like **Microsoft Windows** or **Java**, but without the need for physical infrastructure. Here's how cloud computing is becoming a platform:

1. Understanding a Platform:

- A **platform** is a foundation on which applications run and provide results to users. It can be an operating system, a framework, or a service that supports the execution of programs.
- For example, **Microsoft Windows** is a platform because it allows users to run applications on a personal computer. Similarly, **Java** is a platform because it allows developers to create applications that can run on various devices without needing to be rewritten for each one.

2. Cloud Computing as a Platform:

- In the context of **cloud computing**, the **web itself is becoming a platform**. This means that the cloud is no longer just a place for storing data; it is evolving into an environment where applications are hosted, run, and accessed over the internet.
- Users no longer need to rely on local installations of software (like on a desktop or laptop); instead, they can access and use applications via web browsers, all hosted on the cloud.

- Just like an operating system provides a platform for desktop applications, the cloud now provides a platform for web-based applications, ensuring users have access to them wherever they are.

3. Cloud Computing's Role in Application Delivery (Office 2.0 and More):

- A prominent example of cloud computing as a platform is the shift of **Office 2.0** applications from desktop-based to cloud-based solutions. These include popular office productivity tools such as:
 - **Google Docs**, which allows users to create, edit, and share documents, spreadsheets, and presentations online.
 - **Buzzword**, an online word processor that replicates the functionality of traditional desktop word processors, but entirely within the cloud.
- These tools are now available **through the web** (cloud) rather than being installed locally on a computer, making the cloud a platform for accessing and working with office productivity tools.

4. Advantages of Cloud as a Platform:

- **Access Anywhere, Anytime:** Cloud computing platforms allow users to access applications and data from any device with an internet connection. Whether on a laptop, smartphone, or tablet, the cloud makes it easy to continue work across different devices.
- **Collaboration:** Cloud platforms enable real-time collaboration. Multiple users can access and edit the same document or application simultaneously, making teamwork and sharing information seamless.
- **Scalability:** Cloud platforms provide scalability, meaning resources (like storage or computing power) can be adjusted on demand. As businesses or individuals need more capacity, cloud platforms can provide the resources required without the need to invest in physical infrastructure.
- **Cost-Efficiency:** Since cloud services are often based on a pay-per-use model, users only pay for the resources they use, making it a cost-effective way to run applications and store data.

5. Cloud Platforms in Various Sectors:

- Cloud platforms support not only office productivity tools but also a wide range of other applications, such as:
 - **Software as a Service (SaaS):** Cloud platforms offer entire software packages delivered over the web, such as email, customer relationship management (CRM), and collaboration tools.
 - **Platform as a Service (PaaS):** Developers can use cloud platforms to build, test, and deploy applications without managing the underlying infrastructure.
 - **Infrastructure as a Service (IaaS):** Cloud platforms provide computing resources such as virtual machines and storage, which can be scaled according to the needs of the business or user.

6. Cloud as a Platform for Business and Innovation:

- Businesses are increasingly adopting cloud computing as a platform to host applications and services. This allows companies to reduce hardware and maintenance costs while having access to cutting-edge technologies such as big data analytics, artificial intelligence, and machine learning, all provided through cloud platforms.
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- **Principles of Cloud Computing :**

1. On-Demand Self-Service

Cloud computing allows users to access and utilize resources such as storage, servers and applications without requiring manual intervention from service providers. This means that users can instantly provision the resources they need without waiting for approvals or configurations by IT staff. For example, when signing up for Google Drive, users can immediately start storing and managing files without having to contact Google for setup or assistance.

2. Broad Network Access

Cloud services are designed to be accessible over the internet, allowing users to connect from various devices, including smartphones, laptops, and tablets. This ensures that cloud-based applications and data are available anytime and anywhere, as long as there is an internet connection. A common example is Netflix, which enables users to stream content seamlessly on multiple devices, such as smartphones, smart TVs, or tablets, regardless of their location.

3. Resource Pooling

Cloud providers use a multi-tenant model where computing resources, such as storage and processing power, are dynamically allocated and shared among multiple users. This pooling of resources ensures efficiency while maintaining data security and privacy.

For instance, Dropbox stores files for millions of users on shared cloud servers, but each user's data remains private and isolated from others, ensuring confidentiality and security.

4. Scalability and Elasticity

Cloud computing offers the ability to scale resources up or down based on demand, making it an ideal solution for businesses with fluctuating workloads. This elasticity ensures that users only consume the resources they need at any given time.

For example, an e-commerce website experiencing a surge in traffic during a holiday sale can automatically increase its server capacity to handle the load and then scale down once traffic returns to normal, optimizing performance and cost-efficiency.

5. Measured Service (Pay-as-You-Go)

Cloud computing follows a usage-based billing model where users only pay for the resources they consume, similar to utilities like electricity and water. This approach makes cloud services cost-effective, as businesses and individuals can manage expenses based on actual usage.

A practical example is Amazon S3 cloud storage, where users are billed only for the exact amount of storage they use, such as 50 GB, instead of paying for a fixed, predetermined capacity.

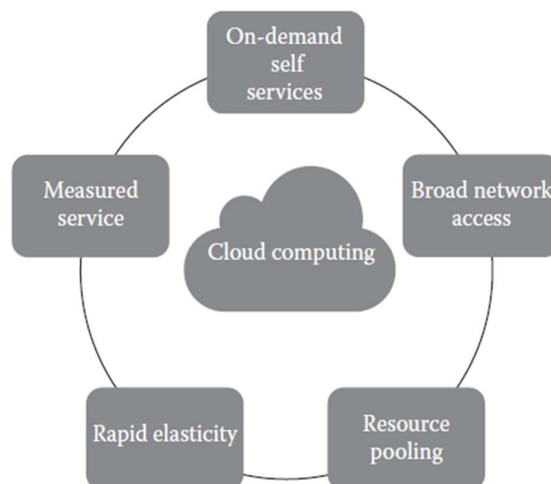
6. Security

Cloud providers implement robust security

measures to protect user data, including encryption, firewalls, authentication protocols, and regular backups. These security mechanisms ensure data integrity and confidentiality, preventing unauthorized access or loss.

For instance, Gmail employs strong encryption and security protocols to protect users email, ensuring that only the intended recipient can access the information, thereby enhancing data privacy and security.

• Five Essential Characteristics :



1. On-Demand Self-Service :

Cloud computing allows users to access services whenever they need them without requiring assistance from IT administrators or service providers. This means users can

quickly set up and manage cloud resources, such as storage, virtual machines, or applications, without waiting for manual approvals or configurations.

For example, when creating an email account on Gmail, a user can complete the entire setup independently without needing to contact a support team. This convenience and automation make cloud services highly efficient and user-friendly, reducing the need for technical expertise.

2. Broad Network Access :

One of the key advantages of cloud computing is its ability to be accessed from anywhere with an internet connection. Cloud services are designed to work on multiple devices, such as smartphones, tablets, and computers, providing seamless accessibility. This ensures that users can work, store data, or access applications remotely without being restricted to a specific location.

For instance, Google Drive allows users to log in from a laptop at home, a desktop at work, or a phone while traveling, making cloud computing a highly flexible and convenient solution for individuals and businesses.

3. Resource Pooling :

Cloud providers use a shared infrastructure where computing resources, such as servers and storage, are dynamically allocated to multiple users. However, each user gets a secure and private environment within the shared infrastructure. This multi-tenant model improves efficiency, as resources are utilized based on demand, reducing wastage.

For example, OneDrive stores files for millions of users on shared cloud servers, yet each user's data remains private and secure, accessible only to them. This approach enables cost-effective and scalable cloud solutions without compromising security or performance.

4. Rapid Elasticity :

Cloud computing offers rapid elasticity, allowing users to quickly scale their resources up or down based on their requirements. This flexibility ensures that businesses and individuals can handle varying workloads efficiently. During peak times, such as a holiday sale, an online store can automatically increase its server capacity to accommodate thousands of visitors. Once traffic decreases, the store can scale down its resources, reducing costs and optimizing performance. This dynamic resource allocation makes cloud computing an ideal choice for businesses with fluctuating demands.

5. Measured Service (Pay-as-You-Go) :

Cloud computing operates on a pay-as-you-go model, where users are charged based on their actual resource consumption. This approach eliminates the need for upfront investments in infrastructure and helps businesses manage costs efficiently. Similar to utility services like electricity or water, users only pay for what they use.

For instance, if someone stores 10 GB of data on Amazon Cloud, they are billed only for that specific usage rather than a fixed storage plan. This cost-effective model makes cloud computing accessible to businesses of all sizes, ensuring that they only pay for the resources they truly need.

These characteristics collectively make cloud computing a powerful solution that is flexible, scalable, and cost-efficient for both individuals and businesses, enabling seamless access to technology without the burden of managing complex infrastructure.

- **Four Deployment Models :**

Q. What is a Cloud Deployment Model ?

A **Cloud Deployment Model** means **how a cloud system is built and used**.

- It explains **where the cloud servers are located** (for example, in a company or with a cloud provider).
- It tells **who owns and takes care of the cloud** (the company itself or a cloud provider).
- It also explains **who is allowed to use the cloud services** (everyone or only specific users).

The type of cloud chosen depends on:

- **Security needs** (how safe the data must be),
- **Access control** (who can use it),
- **Organization needs** (business or institutional requirements).

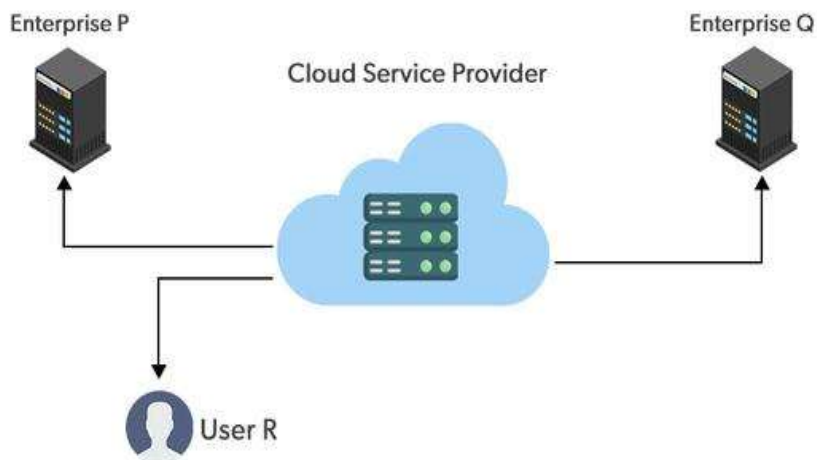
Types of Cloud Computing Deployment Models :

The cloud deployment model identifies the specific type of cloud environment based on ownership, scale and access as well as the cloud's nature and purpose. The location of the servers you're utilizing and who controls them are defined by a cloud deployment model. It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself. Relationships between the infrastructure and your users are also defined by cloud deployment types. Different types of cloud computing deployment models are described below.

- **Public Cloud**
- **Private Cloud**
- **Hybrid Cloud**
- **Community Cloud**

1. Public Cloud :

A **public cloud** is a type of cloud deployment model in which cloud services are made available to anyone through the internet. In this model, the cloud infrastructure is owned, managed and maintained by a cloud service provider, not by the users. Multiple customers share the same resources, which makes the public cloud easy to use and cost-effective, but it is generally considered less secure than a private cloud because it is open to many users. Public cloud services can be accessed for free, through a subscription, or on a pay-per-user basis. This model is widely used because it allows users to access systems and services without worrying about hardware or maintenance. Examples of public cloud services include Google App Engine, Amazon Web Services, and Microsoft Azure.



Public Cloud

Advantages of the Public Cloud Model :

- **Minimal Investment:** Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- **No setup cost:** The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- **Infrastructure Management is not required:** Using the public cloud does not necessitate infrastructure management.
- **No maintenance:** The maintenance work is done by the service provider (not users).
- **Dynamic Scalability:** To fulfil your company's needs, on-demand resources are accessible.

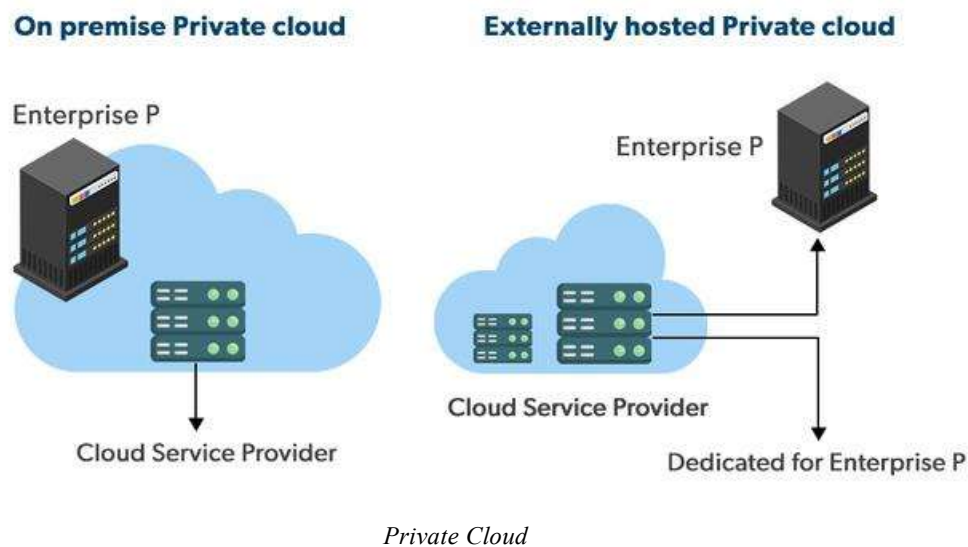
Disadvantages of the Public Cloud Model :

- **Less secure:** Public cloud is less secure as resources are public so there is no guarantee of high-level security.

- **Low customization:** It is accessed by many public so it can't be customized according to personal requirements.

2. Private Cloud :

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between private and public clouds is in how you handle all of the hardware. It is also called the “internal cloud” & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources. Example: Banks using private data centers.



Advantages of the Private Cloud Model :

- **Better Control:** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- **Data Security and Privacy:** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- **Supports Legacy Systems:** This approach is designed to work with legacy systems that are unable to access the public cloud.
- **Customization:** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

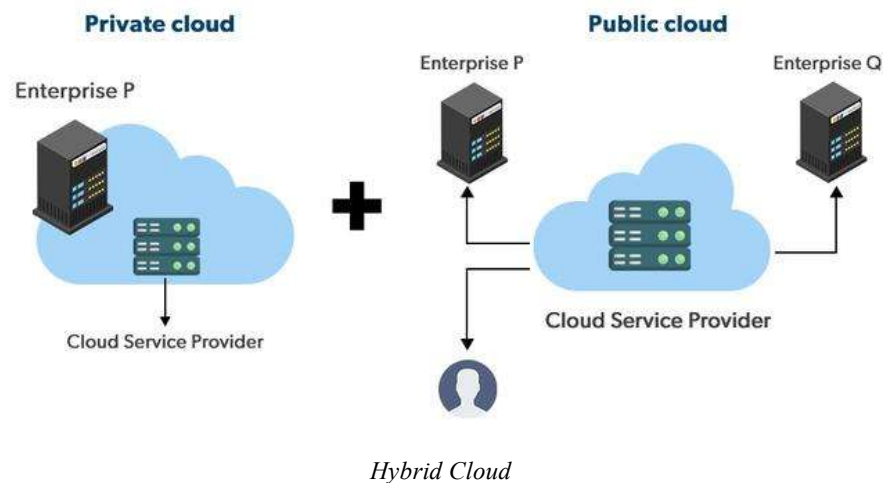
Disadvantages of the Private Cloud Model :

- **Less scalable:** Private clouds are scaled within a certain range as there is less number of clients.

- **Costly:** Private clouds are more costly as they provide personalized facilities.

3. Hybrid Cloud :

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs. Example: A business storing sensitive data on a private cloud while using public cloud for less critical workloads.



Advantages of the Hybrid Cloud Model :

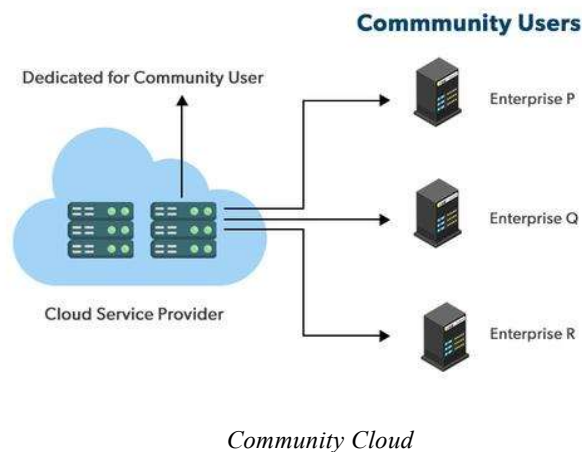
- **Flexibility and control:** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- **Cost:** Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- **Security:** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages of the Hybrid Cloud Model :

- **Difficult to manage:** Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- **Slow data transmission:** Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

4. Community Cloud :

It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community. Example: Government agencies collaborating on a shared platform for public services.



Advantages of the Community Cloud Model :

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model :

- **Limited Scalability:** Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
 - **Rigid in customization:** As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.
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Refer 3 Services of Cloud Computing from Text Book that I have already shared.

❖ Cloud Ecosystem :

What is a Cloud Ecosystem?

A **cloud ecosystem** is like a **teamwork system** where many people, companies, software and hardware work **together** to provide cloud services smoothly.

Just like a **school ecosystem** has teachers, students, books, classrooms and staff.

A **cloud ecosystem** has users, service providers, partners, software, hardware and networks.

All of them depend on each other.

Main Actors in a Cloud Ecosystem:

1. Cloud Service Users (CSUs) – *Users of cloud services*

These are **people or organizations who use cloud services**.

Examples:

- **Individual:**
 - A student using **Google Drive** to store notes
 - A person using **Gmail**
- **Enterprise (Company):**
 - A company using **AWS** to host its website
 - An office using **Tally on Cloud**
- **Government:**
 - Government using cloud for **Aadhaar data** or **online portals**

End users can be:

- **People** (students, employees)
 - **Machines** (IoT devices)
 - **Applications** (apps accessing cloud data)
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2. Cloud Service Providers (CSPs) – *Service givers*

These are **companies that provide cloud services** like software, platform, or infrastructure.

Examples:

- **SaaS (Software as a Service)**
 - Gmail
 - Tally Cloud

- Zoom
- **PaaS (Platform as a Service)**
 - Google App Engine
 - AWS Elastic Beanstalk
- **IaaS (Infrastructure as a Service)**
 - Amazon EC2
 - Microsoft Azure Virtual Machines

CSPs manage:

- Servers
- Storage
- Networks
- Security

Users just **use the service**.

3. Cloud Service Partners (CSNs) – *Supporters and helpers*

These are **third-party people or companies** that help CSPs to **build, improve, or support** cloud services.

Examples:

- **Application developers**
 - Developers creating apps that run on AWS or Azure
- **Content providers**
 - Companies providing data, videos, or reports
- **System integrators**
 - Companies that connect cloud software with existing systems
- **Auditors / Consultants**
 - Security auditors checking cloud safety
 - Consultants helping companies move to cloud

Example: Tally Cloud Ecosystem

- **CSU (User):**
 - A shop owner using **Tally** for accounting
- **CSP (Provider):**
 - Tally company providing accounting software on cloud
- **CSN (Partner):**
 - Another company providing:
 - Reports & dashboards

- CRM
- Payroll tools
- Project management tools

Together, they give **one complete solution** to the customer.

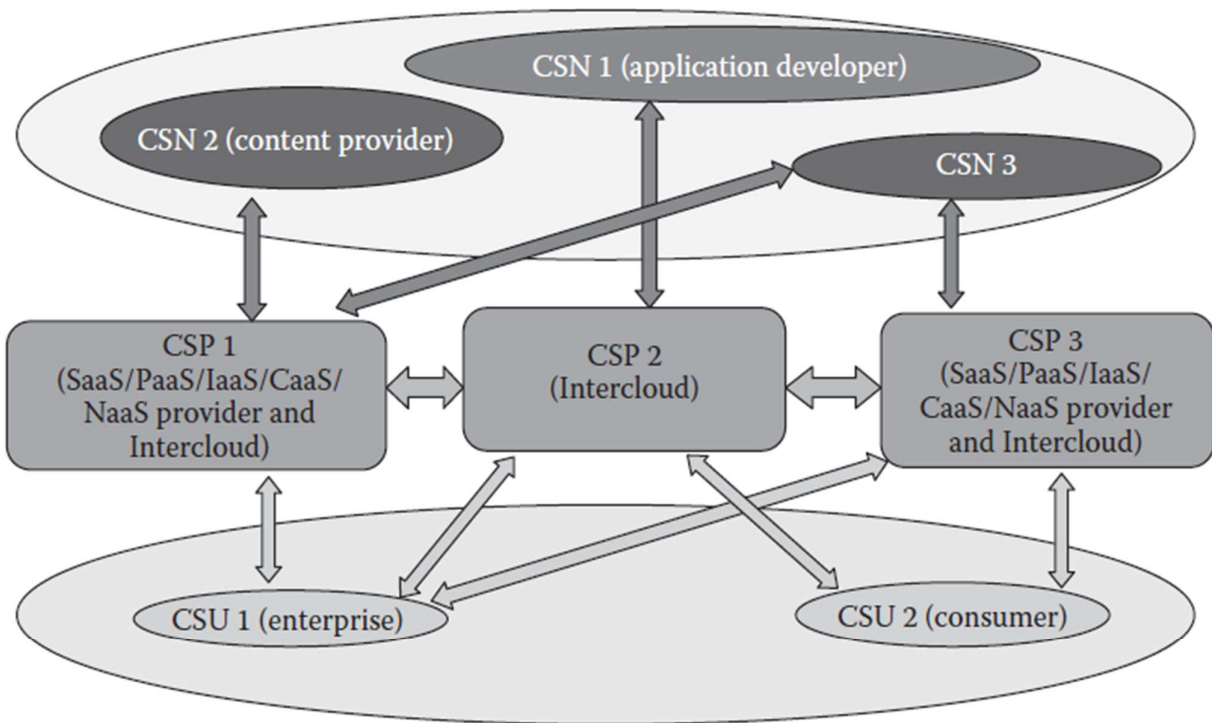


FIGURE 2.4

Actors with some of their possible roles in a cloud ecosystem.

❖ Requirements to be considered for cloud services :

When an organization decides to use cloud services, it cannot adopt them blindly. It must carefully examine and analyse different requirements related to technology, daily operations, and business needs. This evaluation is necessary to ensure that the selected cloud services work efficiently, remain secure, meet organizational goals, and provide value for money.

1. Scalability

Scalability refers to the **ability of cloud services to increase or decrease computing resources automatically** according to workload demands.

In traditional systems, hardware capacity is fixed and expansion requires purchasing new servers, which is time-consuming and expensive. Cloud computing overcomes this limitation by offering **on-demand scalability**. Resources such as CPU, RAM, storage, and network bandwidth can be scaled **vertically** (adding more power to an existing system) or **horizontally** (adding more virtual machines).

Scalability ensures that applications perform smoothly during peak loads without wasting resources during low usage periods.

Example:

During online admission or result declaration, a college website experiences heavy traffic. Cloud services automatically add more servers to handle the load.

2. Availability

Availability is the ability of cloud services to be **accessible and operational at all times**, with minimal downtime.

High availability is achieved through **redundant infrastructure**, multiple data centers, load balancing, and failover mechanisms. If one server or data center fails, traffic is redirected to another working resource automatically. Availability is usually expressed in percentage form (e.g., 99.9%).

Cloud users must evaluate how critical continuous access is to their application and choose a provider that offers suitable availability guarantees.

Example:

Online banking and email services must remain available 24×7 without service interruption.

3. Reliability

Reliability refers to the **consistent and dependable performance of cloud services** over time.

A reliable cloud system ensures that services function correctly even in case of hardware failure, software errors, or network issues. Cloud providers implement **data replication, regular backups, and disaster recovery mechanisms** to maintain reliability.

Reliability ensures that user data is not lost and services continue running smoothly.

Example:

Cloud-based ERP systems must reliably process transactions without data corruption or loss.

4. Security

Security is the protection of **cloud data, applications, and infrastructure** from unauthorized access, attacks, and data breaches.

Cloud security involves multiple layers, including **authentication, authorization, encryption, firewalls, intrusion detection systems, and access control**. Security in cloud computing follows a **shared responsibility model**, where the cloud provider secures the infrastructure and the customer secures data and applications.

Organizations must ensure proper identity management and data protection policies.

Example:

Encrypting student or employee records stored in cloud databases to prevent unauthorized access.

5. Data Privacy and Compliance

Data privacy and compliance ensure that cloud services **follow legal, regulatory, and industry standards** for handling sensitive data.

Organizations must comply with government regulations related to data storage, processing, and access. Some regulations require data to be stored within specific geographic locations. Cloud providers should support compliance standards such as data encryption, audit logs, and access controls.

Failure to comply may result in legal penalties and loss of trust.

Example

Storing healthcare or financial data as per government data protection laws.

6. Performance

Performance refers to the **speed, responsiveness and efficiency** of cloud services.

Cloud performance depends on server capacity, network bandwidth, latency, workload distribution, and virtualization techniques. Cloud providers use **load balancers, high-speed networks, and optimized hardware** to maintain high performance.

Poor performance can affect user experience and application efficiency.

Example:

Video conferencing and streaming applications require low latency and fast data processing.

7. Cost and Pricing Model

Cost refers to the **expense incurred for using cloud services**, based on usage and service type.

Cloud services follow a **pay-as-you-go** pricing model where users pay only for consumed resources. Costs may include compute time, storage space, data transfer, and additional services. Organizations must analyze cost forecasting, budgeting, and hidden costs to avoid overspending.

Proper cost management tools help in optimizing cloud expenditure.

Example:

Paying only for the number of hours a virtual machine is running.

8. Interoperability and Portability

Interoperability is the ability of cloud services to **work with different platforms**, while portability allows **easy movement of applications and data** between cloud providers.

Organizations should avoid vendor lock-in by choosing cloud solutions that support open standards, APIs, and containerization technologies. Portable applications can be migrated without major code changes.

This provides flexibility and long-term cost benefits.

Example:

Moving a web application from a private cloud to a public cloud with minimal configuration changes.

9. Service Level Agreement (SLA)

A Service Level Agreement (SLA) is a **formal contract** between the cloud provider and the user that defines service expectations.

SLAs specify uptime guarantees, performance benchmarks, support response times, data protection policies, and penalties for service failures. Organizations should carefully read SLAs to understand responsibilities and limitations.

A strong SLA ensures trust and accountability between the provider and customer.

Example:

An SLA guaranteeing 99.9% service availability with compensation for downtime.

10. Manageability and Monitoring

Manageability refers to the **ease of controlling, monitoring and maintaining cloud resources**.

Cloud platforms provide dashboards, logs, alerts and automation tools to monitor system health, resource usage, security events, and costs. Good manageability allows administrators to quickly identify and resolve issues.

Automation reduces human effort and operational errors.

Example:

Using monitoring tools to track CPU utilization, storage usage and network traffic.
